The triangle inequality

Core Concepts: The triangle inequality

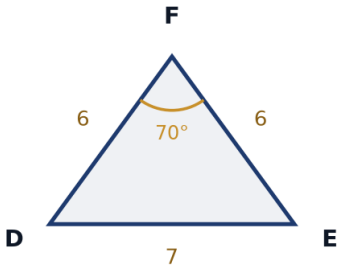
1 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 91°

b. 101°

c. 83°

d. 136°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12

d. 17

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 84°

b. 94°

c. 76°

d. 136°

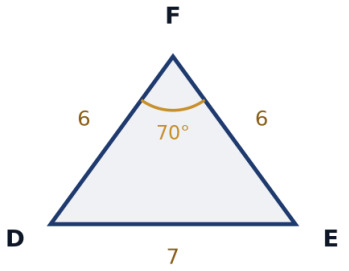
5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm

c. 32 cm

d. 72 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17

d. 22

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 77°

b. 87°

c. 69°

d. 136°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

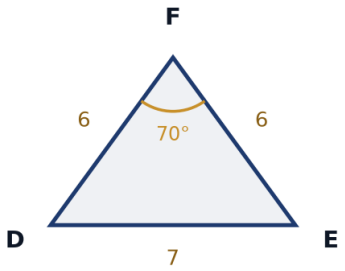
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 17

b. 24

c. 22

d. 27



Exam-Style Problem Solving & Modeling

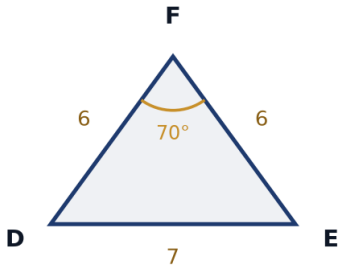
10 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 70°

b. 80°

c. 62°

d. 136°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 22

b. 29

c. 27

d. 32

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